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EfficientAD: Accurate Visual Anomaly Detection at Millisecond-Level Latencies

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Abstract

Detecting anomalies in images is an important task, especially in real-time computer vision applications. In this work, we focus on computational efficiency and propose a lightweight feature extractor that processes an image in less than a millisecond on a modern GPU. We then use a student–teacher approach to detect anomalous features. We train a student network to predict the extracted features of normal, i.e., anomaly-free training images. The detection of anomalies at test time is enabled by the student failing to predict their features. We propose a training loss that hinders the student from imitating the teacher feature extractor beyond the normal images. It allows us to drastically reduce the computational cost of the student–teacher model, while improving the detection of anomalous features. We furthermore address the detection of challenging logical anomalies that involve invalid combinations of normal local features, for example, a wrong ordering of objects. We detect these anomalies by efficiently incorporating an autoencoder that analyzes images globally. We evaluate our method, called EfficientAD, on 32 datasets from three industrial anomaly detection dataset collections. EfficientAD sets new standards for both the detection and the localization of anomalies. At a latency of two milliseconds and a throughput of six hundred images per second, it enables a fast handling of anomalies. Together with its low error rate, this makes it an economical solution for real-world applications and a fruitful basis for future research.

1. Introduction

In the past years, deep learning methods have continued to improve the state of the art across a wide range of computer vision applications. This progress has been accompanied by advances in making neural network architectures faster and more efficient [47, 63, 65, 68]. Modern classification architectures, for example, focus on characteristics such as latency, throughput, memory consumption, and the number of trainable parameters [34, 35, 58, 63, 64, 68]. This ensures that as networks become more capable, their computational requirements remain suitable for real-world applications. The field of visual anomaly detection has also seen rapid progress in the recent past, especially on industrial anomaly detection benchmarks [7, 9, 51, 54]. State-of-the-art anomaly detection methods, however, often sacrifice computational efficiency for an increased anomaly detection performance. Common techniques are ensembling, the use of large backbones, and increasing the input image resolution to up to 768×768 pixels.

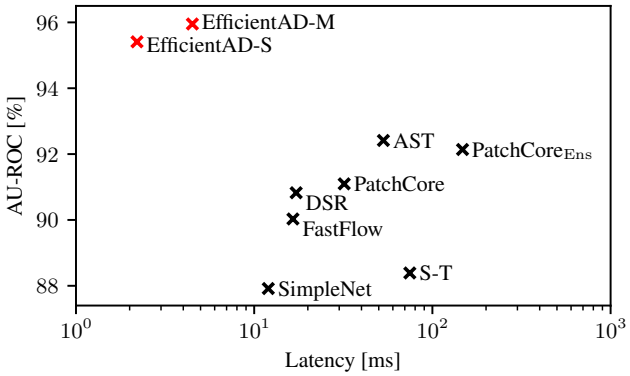


Figure 1. Anomaly detection performance vs. latency per image on an NVIDIA RTX A6000 GPU. Each AU-ROC value is an average of the image-level detection AU-ROC values on the MVTec AD [7, 9], VisA [74], and MVTec LOCO [8] dataset collections.

Real-world anomaly detection applications frequently put constraints on the computational requirements of a method. There are cases where detecting an anomaly too late can cause substantial economic damage, such as metal objects in a crop field entering the interior of a combine harvester. In other cases, even human health is at risk, for example, if a limb of a machine operator approaches a blade. Furthermore, industrial settings commonly involve strict runtime limits caused by high production rates [4]. Not adhering to these limits would decrease the production rate of the respective application and thus its economic viability. It is therefore essential to pay attention to the computational and economic cost of anomaly detection methods to keep them suitable for real-world applications.